

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – IX

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Class: TYIT	Batch: 1
Date of Assignment: 18/8/26	Date/Time of Submission: 18/8/26

a. Develop CRUD operations using ASP.NET Core Web API and Entity Framework Core.

CODE:

StudentController.cs

```

using Microsoft.AspNetCore.Mvc;
using StudentAPI.Data;
using StudentAPI.Models;
namespace StudentAPI.Controllers
{
    [Route("api/[controller]")]
    [ApiController]
    public class StudentsController : ControllerBase
    {
        private readonly AppDbContext _context;

        public StudentsController(AppDbContext context)
        {
            _context = context;
        }

        // GET ALL
        [HttpGet]
        public IActionResult GetStudents()
        {
            return Ok(_context.Students.ToList());
        }

        // GET BY ID
        [HttpGet("{id}")]
        public IActionResult GetStudent(int id)
        {
            var student = _context.Students.Find(id);

            if (student == null)
                return NotFound();
        }
    }
}

```

```
        return Ok(student);
    }

    // INSERT
    [HttpPost]
    public IActionResult AddStudent(Student student)
    {
        _context.Students.Add(student);
        _context.SaveChanges();

        return Ok(student);
    }

    // UPDATE
    [HttpPut("{id}")]
    public IActionResult UpdateStudent(int id, Student student)
    {
        var data = _context.Students.Find(id);

        if (data == null)
            return NotFound();

        data.Name = student.Name;
        data.Age = student.Age;
        data.Course = student.Course;

        _context.SaveChanges();

        return Ok(data);
    }

    // DELETE
    [HttpDelete("{id}")]
    public IActionResult DeleteStudent(int id)
    {
        var student = _context.Students.Find(id);

        if (student == null)
            return NotFound();

        _context.Students.Remove(student);
        _context.SaveChanges();
        return Ok();
    }
}
```

Program.cs

```

using Microsoft.EntityFrameworkCore;
using StudentAPI.Data;

var builder = WebApplication.CreateBuilder(args);
builder.Services.AddControllers();

builder.Services.AddDbContext<AppDbContext>(options =>
    options.UseSqlServer(
        builder.Configuration.GetConnectionString("DefaultConnection")));

builder.Services.AddEndpointsApiExplorer();
builder.Services.AddSwaggerGen();

var app = builder.Build();

app.UseSwagger();
app.UseSwaggerUI();
app.UseHttpsRedirection();
app.UseAuthorization();
app.MapControllers();
app.Run();

```

AppSetting.json

```

{
  "ConnectionStrings": {
    "DefaultConnection":
"Server=(localdb)\MSSQLLocalDB;Database=StudentDB;Trusted_Connection=True;"
  }
}

```

AppDbContext.cs

```

using Microsoft.EntityFrameworkCore;
using StudentAPI.Models;
namespace StudentAPI.Data
{
    public class AppDbContext : DbContext
    {
        public AppDbContext(DbContextOptions<AppDbContext> options)
            : base(options)
        {
        }
        public DbSet<Student> Students { get; set; }
    }
}

```

Student.cs

```
namespace StudentAPI.Models
```

```
{
```

```
    public class Student
```

```
    {
```

```
        public int Id { get; set; }
```

```
        public string Name { get; set; }
```

```
        public int Age { get; set; }
```

```
        public string Course { get; set; }
```

```
    }
```

```
}
```

OUTPUT:

Swagger
Supported by SMARTBEAR

Select a definition **StudentAPIWithEF v1**

StudentAPIWithEF 1.0 OAS 3.0

<https://localhost:44344/swagger/v1/swagger.json>

Students

- GET /api/Students
- POST /api/Students
- GET /api/Students/{id}
- PUT /api/Students/{id}
- DELETE /api/Students/{id}

WeatherForecast

- GET /WeatherForecast

POST /api/Students

Parameters

No parameters

Request body **application/json**

Edit Value | Schema

```
{
  "name": "Soham",
  "age": 20,
  "course": "BSC IT"
}
```

Execute Clear

Responses

Curl

```
curl -X 'POST' \
  'https://localhost:44344/api/Students' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "name": "Soham",
    "age": 20,
    "course": "BSC IT"
  }'
```

Request URL

https://localhost:44344/api/Students

Server response

Code Details

200

Response body

```
{
  "id": 1,
  "name": "Soham",
  "age": 20,
  "course": "BSC IT"
}
```

Download

Response headers

```
content-type: application/json; charset=utf-8
date: Tue, 18 Aug 2026 12:37:28 GMT
server: Microsoft-IIS/10.0
x-powered-by: ASP.NET
```

Responses

Code	Description	Links
200	OK	No links

GET /api/Students/{id}

Parameters

Cancel

Name Description

id * required
Integer(\$int32)
(path)

1

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'https://localhost:44344/api/Students/1' \
  -H 'accept: */*' \
```

Request URL

https://localhost:44344/api/Students/1

Server response

Code Details

200

Response body

```
{
  "id": 1,
  "name": "Soham",
  "age": 20,
  "course": "BSC IT"
}
```

Download

Response headers

```
content-type: application/json; charset=utf-8
date: Tue, 18 Aug 2026 12:39:17 GMT
server: Microsoft-IIS/10.0
x-powered-by: ASP.NET
```

Responses

PUT

/api/Students/{id}

Parameters

Cancel

Reset

Name	Description
id * required	
Integer(\$int32)	1
(path)	

Request body

application/json

Edit Value | Schema

```
{  "id": 0,  "name": "Soham Acharekar",  "age": 21,  "course": "BSC IT"}
```

Execute

Clear

DELETE

/api/Students/{id}

Parameters

Cancel

Name	Description
id * required	
Integer(\$int32)	1
(path)	

Execute

Clear

Responses

Curl

```
curl -X 'DELETE' \  https://localhost:44344/api/Students/1 \  -H 'accept: */*' \  -H 'Content-Type: application/json' \
```

Request URL

https://localhost:44344/api/Students/1

Server response

Code	Details
200	Response headers
	<pre>date: Tue, 18 Aug 2026 12:46:57 GMT server: Microsoft-IIS/10.0 x-powered-by: ASP.NET</pre>

Responses

Code	Description	Links
200	OK	No links

dbo.Students [Data] 202608181234...tialCreate.cs StudentsCo

Max Rows: 1000

	Id	Name	Age	Course
	2	Soham	20	BSC IT
	3	Soham Acharek...	21	BSC IT
	NULL	NULL	NULL	NULL
	NULL	NULL	NULL	NULL

b. Create an API endpoint to calculate Factorial of a Number and test it using Postman.**CODE:MathController.cs**

```

using Microsoft.AspNetCore.Mvc;
namespace FactorialAPI.Controllers
{
    [Route("api/[controller]")]
    [ApiController]
    public class MathController : ControllerBase
    {
        [HttpGet("factorial/{number}")]
        public IActionResult GetFactorial(int number)
        {
            if (number < 0)
            {
                return BadRequest("Factorial not possible for negative numbers.");
            }

            long factorial = 1;

            for (int i = 1; i <= number; i++)
            {
                factorial *= i;
            }
            return Ok(new
            {
                Number = number,
                Factorial = factorial
            });
        }
    }
}

```

OUTPUT:(POSTMAN)

The screenshot shows the Postman interface. At the top, a GET request is configured to `https://localhost:44352/api/math/factorial/5`. The 'Params' tab is selected. Below, the 'Query Params' table is empty. The response section shows a status of `200 OK` with a response time of `128 ms` and a body size of `196 B`. The response body is displayed in JSON format:

```

{
  "number": 5,
  "factorial": 120
}

```